Triangle geometries of finite linear spaces

Human Compression project

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Abstract

We recover the principal classification of Delaby, Leemans and Tranchida [DLT] through the flag-relation and frame-orbit interface [FR]. The geometric construction, its principal classification target, and the older classification of locally two-transitive linear spaces are explicitly attributed. The group-action statement requires a correction: in an affine plane, semilinear frame transitivity is a condition on determinant orbits, not containment of the full linear group. We give a uniform necessary and sufficient criterion and a concrete counterexample over the field of nine elements. This correction does not change the classified geometries whose triangle systems are firm, residually connected and flag-transitive. The full automorphism and correlation groups, the thin cases, and all diagrams in the principal finite family are determined. The Hermitian connectivity argument reuses an independently reviewed norm-shell theorem; the finite exceptional residue is checked directly from its defining design.

1. Exact scope and retained inputs

Our source edition is the final journal article [DLT], Algebraic Combinatorics 9(2) (2026), 379–402, not its earlier preprint. Its exact PDF is bound in the accompanying source contract. The principal targets are Theorems 1.1 and 1.2, together with the construction, automorphism, correlation and residue statements needed for them and the diagrams and Table 1. We correct literal defects rather than treating a false statement as a theorem we have reproduced. The source's final experimental questions are a separate track and are not counted as part of this recovery. The Desargues-configuration and square examples, and the identification with a numbered hypermap in an external catalogue, are auxiliary assertions outside this principal contract. We do not describe this as recovery of every assertion printed in the paper. The general isomorphism statements between two uniform input spaces are, however, included in [FR, Theorem 2].

We use [FR] for the full flag relation, reconstruction, correlation and residue interface, and its determinant-orbit theorem. These are complete proofs in an actual separate paper. The following retained results remain substantial mathematical inputs.

1. Buekenhout, Dehaye and Leemans [BDL, Theorem 6.2, pp.42–43] classify finite faithful flag-transitive linear spaces whose point stabilizers and line stabilizers act two-transitively on their respective pencils and lines. The classification ultimately uses finite group classification. We retain its necessity list, including its specified natural actions. We check the extra noncollinear-triple condition below; we do not retain an unchecked blanket converse. Its circle paragraph already assumes three-transitivity.
2. The fundamental automorphism theorems for finite projective and affine spaces identify their full groups as PΓL and AΓL, respectively. For the classical Hermitian unital, the full group is PΓU(3,q), with U_H(2) identified with AG(2,3). These classical identifications are retained as in DLT; they are not new group-classification theorems proved here.
3. The complete Hermitian transport and generation proofs are in Part III of the common paper [FR]. Theorems H.1–H.3 prove that a nonzero norm shell is a regular SU(2,q)-set and that two specified

isotropic directions suffice for connectedness. These arguments are included in the present topic, rather than imported without their proof.

4. The usual Mathieu permutation actions and the classical classification of finite three-transitive permutation groups, as retained in BDL's circle case, are used with their actual three-transitivity hypothesis. In its projective-line branch the required natural-action normalizer identity is proved separately in [FR, Theorem 6], including q=2,3; it is not inferred from the dimension-at-least-two collineation theorem. The exceptional A7 linear action needed here is also checked directly in the finite certificate.

All remaining implications in the principal contract are proved below or in [FR]. A computation for finitely many fields is never substituted for an assertion for every prime power.

2. Corrected classification of the actions

For a linear space S, let T(S) be its ordered noncollinear triples. A circle K_v means the linear space whose lines are all two-element subsets of its v-point set. We exclude the degenerate one-line space, consistently with DLT's requirement that at least two lines pass through each point.

Theorem 1 (corrected action classification). Let G act faithfully on a finite linear space S and transitively on T(S). Then (S,G) belongs to one of the following families, and every specified pair has this transitivity.

(a) S=PG(d,q), d≥2, and PSL(d+1,q)≤G≤PΓL(d+1,q).

(b) S=PG(3,2), with the exceptional natural group G≅A7.

(c) S=AG(d,q), d≥2 and q≥3, and G=K^d⋊H with SL(d,q)≤H≤ΓL(d,q). For d≥3 no further restriction is needed. For d=2 the determinant-semilinear image of H must be transitive on K*, under t↦det(A)σ(t) for the transformation v↦v^σ A.

(d) S=U_H(q), q=2 or 4, and G=PΓU(3,q).

(e) S=K_v and G is a finite three-transitive permutation group in its given v-point action. Its customary list is: S_v for v≥3; A_v for v≥5; groups between PSL(2,q) and PΓL(2,q) in their projective-line action which are three-transitive; the Mathieu actions M11,M12,M22,M23,M24 of the indicated degrees and Aut(M22); the degree-12 action of M11; AGL(d,2), d≥2; and the exceptional 2^4:A7 action of degree 16. In the projective-line case every intermediate group is three-transitive when q is even. For odd q its projective determinant-square image must be nontrivial. Coincidences between entries are permitted. There is no A7 example of degree 15.

Proof of necessity. Triple transitivity first implies flag transitivity and the local two-transitivities required by BDL. To see flag transitivity, extend a flag (p,L) to (p,a,b) with a∈L{p} and b outside L, and use the triple action. For two distinct lines through p, choose a and b on them other than p. Sending the resulting triple to any similarly chosen triple fixes p and sends the ordered pair of lines as desired. For two distinct points on L, choose a third point outside L; the same argument gives the two-transitivity of the setwise line stabilizer on L. Faithfulness is part of our hypothesis, so all the assumptions of BDL's classification apply.

Besides (a)–(e), BDL has the exceptional pair AG(2,4), AΓL(1,16). Here there are 16·15·12=2880 ordered noncollinear triples but the acting group has order 16·15·4=960; it cannot be transitive. In the affine family BDL provides the full translation subgroup and the contained SL group. The ambient upper bound is ΓL by the affine automorphism theorem. The remaining condition is exactly [FR, Theorem 5], which proves (c), including the semilinear plane case. The projective and A7 cases need no additional necessity restriction. The unital branch is determined in §3. A circle has no collinear triple of distinct points, so its condition is precisely three-transitivity. The circle list is the retained permutation classification, with the

explicit impossible low-degree entries removed. For its PSL-containing natural projective-line branch, the normalizer identity [FR, Theorem 6] supplies the stated PΓL upper bound. This proves necessity, subject only to the unital calculation supplied next.

Proof of sufficiency outside the unital case. In PG(d,q), three noncollinear points are represented by three independent vectors. Extend them to a vector basis, and do the same for a target triple. A linear map between these bases carries the projective triple correctly. Rescale one target basis vector to make its determinant one; rescaling does not change that target projective point. Hence PSL acts transitively. For (b), the certificate realizes GL(4,2) on its 15 nonzero vectors, identifies its exceptional A7 subgroup, and checks its single orbit of length 15·14·12=2520 on independent ordered triples. This is the complete one-field exception, not a sample of an infinite family. For (c), translating the first point to zero and applying [FR, Theorem 5] proves sufficiency.

For the circle case, three-transitivity is exactly the desired assertion. The standard actions in the list are retained as stated, while the bounds on alternating groups are immediate: S_v sends every ordered triple to every other one; when v≥5 its parity can be corrected on two unused points, so A_v does as well. The orders of A3 and A4 are insufficient. On the projective line PGL(2,q) is sharply three-transitive. For odd q PSL has two orbits on ordered triples, distinguished by the determinant-square class of the unique projectivity carrying (∞,0,1) to that triple. A field automorphism fixes the base triple and preserves square classes. Thus an intermediate semilinear group is transitive exactly when it has a nonsquare-determinant element. For even q PSL=PGL. In affine binary space, any two distinct nonzero vectors are independent; GL(d,2) is transitive on such pairs, proving the assertion for AGL(d,2). The finite certificate also checks that the same exceptional A7 in GL(4,2) is two-transitive on the 15 nonzero vectors, which proves the 2^4:A7 assertion after translations. The standard Mathieu actions are retained, as declared above. ■

The impossible A7 circle entry already occurs in the older BDL display and was already identified in the project's earlier [IC, §4]. Its order 2520 is smaller than 15·14·13=2730. This observation does not affect its valid PG(3,2) action, whose noncollinear count is2520. DLT's named circle list also permits A_v for v=3,4; its preliminary three-transitivity hypothesis excludes those groups, but the displayed named converse must not be read as asserting their three-transitivity. These are reused source-audit findings, not newly answered research questions.

3. The unital orbit parameter

Use homogeneous coordinates over K=F_(q²) and the Hermitian form

$$h(x,x) = x_1x_3^q + x_1^q x_3 + x_2^{q+1}.$$

The projective unitary group is two-transitive on the isotropic points; this is a standard consequence of Witt extension, retained here. Fix an ordered pair as e1,e3. Every isotropic point outside their joining line has a unique representative (x1,1,x3), with both x1 and x3 nonzero, and

$$z = x_1x_3^q, \qquad z + z^q = -1. \tag{9}$$

The complete PGU stabilizer of e1,e3 is represented by diag(a,1,a^(−q)), a∈K*. Indeed preservation of the two isotropic lines also preserves their orthogonal complementary line; after projective normalization of its middle diagonal entry the Hermitian condition gives exactly these matrices. They preserve z and act regularly on each z-fiber: normalize x1 to1 and then x3 is fixed by (9). In particular there are q PGU orbits on T(S), and the stabilizer of any ordered noncollinear triple is trivial.

The field group acts on z by its p-power Frobenius, where q=p^e. If q is odd, z=−1/2 is a fixed member of (9), while the fiber has q>1 elements. Thus no subgroup of PΓU is transitive. If q=2^e with e≥3, every field orbit has length at most2e<q, again excluding transitivity. For q=2 the trace-one fiber consists of the two elements of F4\F2, a single Frobenius orbit. For q=4 it has four elements, none in F4, since an element of F4 has trace z+z^4=0. They all have degree four over F2 and form one Frobenius orbit. The full PΓU group is therefore transitive exactly for q=2,4.

In both cases its order equals |T(S)|:

$$|T(S)| = (q^3+1)q^3(q^3-q), \qquad |\,\mathrm{P\Gamma U}(3,q)| = (q^3+1)q^3(q^2-1)\,2e.$$

Here q=2e for q=2,4. Any transitive subgroup has order at least |T(S)|, so it must be the full group. This proves both directions of Theorem 1(d), including the group requirement that cardinal inequalities alone do not supply for arbitrary q. The small identification U_H(2)≅AG(2,3) is also verified by the finite model; it is not used to infer the q=4 calculation.

4. A semilinear counterexample and the repaired affine statement

Take K=F9, let z generate K*, and let σ:x↦x³ be its nontrivial field automorphism. Define H≤ΓL(2,9) by the condition

$$(A,\sigma^e) \in H \quad \Longleftrightarrow \quad \det A \text{ is a square exactly when } e=0, \qquad e \in \{0,1\}. \tag{10}$$

This is a subgroup: Frobenius preserves square classes and the determinant class of a composite is the product of the two determinant classes. Its linear half contains SL(2,9), of order 720, and has determinant image the four squares. The other half is obtained using (diag(z,1),σ). Thus |H|=2·4·720=5760. It omits every nonsquare-determinant linear matrix and therefore does not contain GL(2,9), which also has order 5760.

The induced action (8) is transitive on K*: the linear half is transitive within either square class, and the specified semilinear element interchanges the two classes. By [FR, Theorem 5], H is transitive on the 5760 ordered bases of K², hence regular there. Adding all translations gives a group of order81·5760=466560 acting regularly on the ordered noncollinear triples of AG(2,9). These transformations preserve every affine line, so this is a counterexample within the exact hypotheses of DLT Theorem 1.1, not merely an abstract permutation action. The native GAP certificate verifies the defining line action, subgroup orders, noncontainment and the entire triple orbit.

The replacement in Theorem 1(c) is uniform for all finite fields and all semilinear subgroups containing SL. If one adds the extra assumption that H is linear, it reduces to the original GL conclusion. This distinction is a source correction and a positive orbit theorem; it is not an answer to an explicitly open conjecture.

5. Residual connectedness and the classified geometries

Theorem 2 (triangle classification). For a finite linear space S, Δ(S) is firm, residually connected and flag-transitive if and only if S is PG(2,q), q≥2; AG(2,q), q≥3; or U_H(q), q=2 or4. In each of these cases Δ(S) has a duality exactly when S does.

Suppose first that Δ(S) has the three stated properties. Flag transitivity forces uniformity by [FR, §1], and firmness forces k≥3. Moreover r≥3: if r=2, v=2k−1 and b=vr/k=4−2/k would be an integer; k≥3 is impossible. Consequently [FR, Theorem 2] reconstructs the full automorphism group of S, and its chamber correspondence puts S in Theorem 1. The affine and projective cases of dimension at least

three have disconnected vertex residues: [FR, Theorem 4] separates them into the distinct planes through a fixed line. Circles have k=2 and are not firm. The remaining list is exactly the one stated.

We prove that all surviving cases have the required properties. The appropriate groups in Theorem 1 give flag transitivity; k≥3 gives firmness. Global connectivity follows from [FR, §3]. It remains to show that every vertex residue is connected. It suffices to check the deleted-pencil model [FR,(7)], either uniformly at every flag or at one flag when its group is already flag-transitive.

In a projective plane, deleting the line L leaves an affine plane. The allowed pencils in (7) are all its parallel classes except the class indexed by p. Choose two remaining classes. Every line of one meets every line of the other, so their union and all outside points form a connected incidence grid. Every additional allowed line meets this grid. This proves residue connectivity, including order2.

In an affine plane of order q≥3, choose two distinct a,b in L{p}. There are q non-L lines through each. A line through a meets every one through b except its unique parallel, and those meetings are outside L. The intersection graph of the two pencils is K_(q,q) minus a perfect matching, which is connected for q≥3. Each outside point lies on exactly one line from each pencil, and each additional allowed line contains outside points. This proves the full residue connected. The argument requires no Desarguesian coordinates.

For a Hermitian unital, apply Theorems H.1–H.3 of the common paper. With L fixed, its outside isotropic points identify with a nonzero norm shell in a Hermitian two-space, and its points on L identify with isotropic directions. An edge through a point a of L is the corresponding isotropic-direction move. Formula (7) deletes precisely the direction p. There are q+1 isotropic directions, so for q≥2 choose two distinct allowed directions. Choose representatives e,f of those lines. Their Hermitian pairing is nonzero, and rescaling gives a hyperbolic basis with those two axes. Thus the two explicit root families of [FR, Part III] apply to these particular allowed directions. They generate SU(2,q); by norm-shell transitivity, the moves in these allowed directions connect every outside point. Every allowed-line vertex contains outside points. Thus the entire residue is connected. This proves the stronger uniform result that Δ(U_H(q)) is firm and residually connected for every prime power q≥2, although it is flag-transitive only for q=2,4. The full transport, root generation and geometric identification are proved in [FR, Part III]. The allowed directions have been specified explicitly; the argument does not assume that deleting an arbitrary part of a connected graph preserves connectedness.

Finally all surviving spaces are thick as rank-two spaces. The duality assertion is [FR, Theorem 3]. This completes the proof of Theorem 2. ■

This proves DLT's geometric principal theorem without changing its list. In particular AG(2,q) for q>3 yields a thick triangle geometry with a triality and no duality: it has q^2 points and q(q+1) lines, precluding a duality of its input. U_H(4) also has unequal point and block counts65 and208. Projective planes have their usual polarities.

6. The complete residue diagrams

Write k for the input line size. Each type contains vr flags, and every panel has k−1 completions. Consequently the vertex labels in the usual Buekenhout convention are s=k−2 and n=vr. Triality identifies the three residue types. We give girth, both part-diameters, and all small exceptions.

For PG(2,q), the residue is the affine plane with one parallel class removed. When q=2 it is the eight-cycle. For q≥3 its girth is6 and both part-diameters are4. Indeed any two points either have an allowed joining line or are joined by a four-step path using two allowed directions. A point and a nonincident allowed line have distance3, using another allowed direction through the point. Distinct parallel allowed lines have distance4 through a line of another class, and nonparallel lines have distance2. There are at least three allowed directions; choose a line in each with the third avoiding the intersection of the first

two. Their three pairwise intersections give a six-cycle. No four-cycle can exist in a linear space. Thus every edge label is (4,3,4), except that q=2 gives a generalized quadrangle of order(1,1), with label4. The vertices have n=(q²+q+1)(q+1) and s=q−1.

For AG(2,q), put L={y=0}, p=(0,0). Its residue has outside points (x,y), y≠0, and lines represented by (a,t), a≠0, t∈K, with incidence

$$x = a + ty. \tag{11}$$

For q=3, there are two allowed anchors a, and the graph is the subdivision of K_(3,3) minus a perfect matching: a twelve-cycle. For q≥4, every two point vertices have distance at most4. To see this when there is no common allowed line, choose a line through the first point with anchor a. Among the q−2 allowed anchors b≠a for lines through the second point, at most one yields a parallel line. At least one other choice remains, and its intersection with the first line is outside L. This supplies the desired four-step path. For two line vertices lacking an outside intersection, choose an outside point Q on the first. Of the q−1 outside points on the second, at most one has the same y-coordinate as Q and at most one joins Q to p. Any other choice gives an allowed joining line and a four-step path. Similarly a point and a nonincident allowed line have distance3: choose an anchor different from the line's anchor and avoid the at most one parallel choice. These counts hold already for q=4.

There exist point pairs on a horizontal line whose joining line is not allowed, and two distinct residue lines through the same allowed anchor; both exhibit distance4. There are no four-cycles. To obtain a six-cycle, choose two nonparallel allowed lines with distinct anchors, then a third anchor distinct from both. Of the q lines at that anchor, at most two are parallel to the first pair and at most one passes through their outside intersection. Since q≥4 a remaining line forms a triangle wholly outside L. Thus the girth is6 and both part-diameters are4. Every edge label is (4,3,4), except for q=3 where it is the generalized-hexagon label6. The vertices have n=q²(q+1) and s=q−2.

For U_H(2), use its identification with AG(2,3). For U_H(4), construct the 65 isotropic points in PG(2,16) and its208 secant lines directly. Delete one pencil exactly as in (7). The resulting graph has60 vertices in each part, degree 4, and240 edges. The certificate performs breadth-first search from every vertex, finding the following complete distance profiles:

Part	Numbers of vertices at distances 0,1,2,3,4,5,6
outside-point vertices	1,4,12,36,44,20,3
allowed-line vertices	1,4,12,36,42,20,5

The same searches give shortest circuit length 8. Hence both diameters are 6 and the gonality is 4. Moreover these two different profiles are automorphism invariants, so no graph automorphism interchanges the parts: the rank-two residue has no duality. The full group is flag-transitive by §3, so the calculation at one flag gives every vertex residue, and triality gives all type pairs. The three diagram edges are (6,4,6), and the vertex labels are n=65·16=1040 and s=3. This is a complete bounded certificate for the only exceptional diagram, with no group-order guess or sampled-vertex inference.

7. Full groups, higher dimensions and the thin cases

For all uniform thick cases the exact correlation description is C3⋊Cor(S), with the sign action in [FR, Theorem 3]. The classical full automorphism identifications in §1 therefore give the following table. The symbol "thin" refers to Δ, not to its rank-two input.

Input S	Connected	Residually connected	Thin	Aut(Δ)	Cor(Δ)
AG(d,q), d≥2,q≥3	yes	exactly d=2	exactly q=3	AΓL(d,q)	AΓL(d,q)×C3
PG(d,q), d≥2	yes	exactly d=2	exactly q=2	PΓL(d+1,q)	see below
U_H(4)	yes	yes	no	PΓU(3,4)	PΓU(3,4)×C3
K_v, v≥4	yes	no	no	S_v	S_v×S3
K_3	no	no	no	S6	S6×S3

For PG(2,q), its standard polarity gives Cor(Δ)≅PΓL(3,q)⋊S3; here C3 centralizes the collineation group and the transposition acts by that polarity. For d≥3 there are different numbers of points and lines, so Cor(Δ)=PΓL(d+1,q)×C3. This also supplies the higher-dimensional q=2 rows that are not all printed in DLT's table. The U_H(2) row duplicates AG(2,3). The affine rows require AΓL, including field automorphisms. For a direct witness when q is not prime, Frobenius preserves every affine line but cannot be K-affine linear: it fixes 0 and the standard basis, while moving z e1 for a field element z outside the prime field. In particular the literal AGL label in DLT Table 1 omits actual automorphisms.

For completeness we prove all circle statements. Theorem 2 of [FR] applies when v≥4, giving Aut(Δ(K_v))=S_v. The map

$$(p, \{p, q\}, i) \longmapsto (q, \{p, q\}, -i) \tag{12}$$

is an incidence-preserving involution: it reverses the endpoint order of each directed edge and reverses the cyclic type order. Together with τ it is a copy of S3, commuting with the diagonal S_v. Every possible type permutation is represented and the kernel is exactly S_v, proving Cor(Δ)=S_v×S3.

Its underlying directed flag relation is the nonbacktracking arc relation on K_v: (q,p) advances to (p,a) with a distinct from q,p. For v≥4 its arc graph is strongly connected. For example an arc can be reversed by the walk q,p,a,b,p,q, using two additional distinct vertices, after which the ordinary connectedness of K_v permits movement to any arc without an immediate reversal. Based at any arc there are directed cycles of lengths three and four. Lifting to the cyclic type coordinate, the four-cycle changes the type by one, while returning to the same arc. Thus the lift Δ is connected. In its vertex residue, (7) reduces to v−2 disjoint edges: an outside point Q is incident only to the allowed line qQ, where L={p,q}. It is not residually connected and its panels have only one completion, so it is neither firm nor thin. When v=3, every flag extends to exactly one chamber and Δ is the disjoint union of its six chambers. Its type-preserving automorphisms permute those chambers arbitrarily, giving S6; the additional type permutations give the direct product S6×S3. This proves the entire small-case table without applying a thick reconstruction theorem outside its hypotheses.

8. Literal source corrections and proof coverage

The principal geometry classification, construction, triality and source authorship remain unchanged. The corrected action classification replaces one affine-plane necessity by the exact determinant criterion. The finite circle-list defects were already recorded in [IC]; no new question credit is attached to them. The affine full-group table includes field automorphisms. The general correlation formula C3⋊Cor(S) does not require a polarity, whereas splitting its natural quotient onto S3 does.

There is one further logical limitation in DLT's description of its construction as a functor. Its §2 morphisms need preserve incidence only in the forward direction, while Construction 1 uses both incidence and nonincidence. Such a morphism does not in general induce a triangle morphism. Indeed map every input point to a fixed point p and every line to a fixed incident line L. This is a type-preserving incidence morphism, but its proposed triangle map sends each type to (p,L,i), and no such vertices of different types

are incident. This loses every source edge. The construction does transport correlations, as proved in [FR], and more generally maps preserving both incidence and nonincidence. No use of the false general functor assertion enters either principal proof.

The Lean file proves actual equal-determinant matrix transport, semilinear determinant transformation and subgroup orbit saturation. It does not formalize BDL's classification, the finite geometry model, all residue connectivity, or all source theorem instances. The finite certificate proves the complete fixed exceptional models and advertised controls, not the all-field classification on its own. The complete written proof consists of this application, its full [FR] dependency and the expressly retained inputs. A future matched compression measurement must charge all of them under the agreed topic accounting rule. No ratio, principal-source admission or open-question count is inferred from the number of lemmas or from a successful executable replay.

References

[DLT] R. Delaby, D. Leemans and P. Tranchida, *Geometries with trialities arising from linear spaces*, Algebraic Combinatorics 9(2) (2026),379–402, DOI10.5802/alco.482.

[BDL] F. Buekenhout, P.-O. Dehaye and D. Leemans, *RWPRI and (2T)1 flag-transitive linear spaces*, Beiträge zur Algebra und Geometrie44(1) (2003),25–46. Theorem 6.2, printed42–43, is a retained classification.

[FR] *Flag relations and derived incidence geometries*, the common paper of this topic. Numbered results 1–6 are in Part I; results H.1–H.3 are in Part III.

[IC] *Incidence compression*, earlier repository manuscript, §4. Its matrix reconstruction, correlation extension, residue bridge and already recorded source corrections are reused with attribution.

[HU] The project's earlier proof for Hermitian unital residues and its independent review. Its complete transport and generation arguments are reproduced in [FR, Part III], with the geometric identification for the triangle construction.